# GeoIntel: Final Project Report

Generative AI and Social Media

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# Executive Summary

**Repository URL:** https://github.com/Ben-W-Yal/Good_Vibes_Final_Project

**Main takeaway:** Combining interactive geospatial visualization with citation-grounded generative briefs helps users move faster from raw headlines, wire copy, and **social-media-surfaced events** to defensible written assessments tied to place and URLs.

**Who this is for:** Students, conflict and OSINT observers, journalists, and technically minded analysts who need one workspace to relate fast-moving signals—including those that first break on social platforms—to geography and evidence.

GeoIntel is a geospatial intelligence workstation built with TypeScript/React, Express, and Gemini-powered generative AI. It fuses open-source and news-indexed signals (many of which originate as or are amplified by **social posts**) with tracker overlays and structured AI outputs so users interpret what is happening, why it matters, and what to watch next. Instead of passively scanning disconnected feeds, users work on an interactive 3D globe with optional terrain and buildings (Cesium ion), event filters, and briefing panels.

The interface pairs narrative intelligence with geographic grounding: grounded web research surfaces article imagery where available; users pan, tilt, and zoom through the environment where events unfold. Selecting an event opens a streamed **AI analyst summary** and optional **chat** scoped to that record. That combination supports a clearer mental map of borders, urban fabric, and coastlines—relating headlines and viral narratives to real geography, not abstract place names alone.

**Design rhythm:** the UI keeps stable geographic and feed context while users add refreshed markers, regenerated PDB sections, and per-event summaries. That mirrors manual analysis: maintain a working picture, update when credible signals arrive, without claiming a formal latent-state model in code.

# 1 Introduction

## 1.1 Motivation

Understanding modern conflict increasingly means making sense of fast-moving, fragmentary signals: **social-media posts** (often the first ripple around protests, strikes, or disasters), wire copy, tracker data, and satellite news that rarely arrive in one place. Real-time social reporting can surface genuine eyewitness detail; it can also spread rumor, coordinated inauthentic behavior, and context-stripped clips. Consumers and analysts face the same failure mode: headlines without terrain, timelines without geography, and narratives that age out before context arrives. GeoIntel addresses this gap by coupling those signals—as they appear in our normalized event feeds—with interactive mapping and generative summarization so users relate events to *where* they occur and *what* surrounds them—terrain, urban structure, coastlines, and borders—rather than scanning disconnected feeds.

### 1.2 Problem statement and intended users

The core problem is **situational overload**: too many sources (including high-volume **social-derived** and news-indexed events), too little integration, and weak coupling between narrative reporting and geographic reality. GeoIntel targets students, analysts, journalists, and technically minded observers who need a single workspace to explore conflict-related signals, overlay mobility where available, and produce structured written assessments grounded in web evidence.

### 1.3 Plain-language system description

**Inputs** include configurable OSINT-style event feeds (e.g., GDELT-style document and media intensity [4], which indexes global news and **social-linked** discussion, plus structured conflict datasets such as ACLED [5] where enabled), optional live trackers for aircraft, maritime traffic, and satellite propagation from public catalogs [3], plus user-selected filters for region and time.

**Outputs** include an interactive **3D globe** (CesiumJS [2]) with terrain and building visualization where configured. Briefing panels incorporate grounded article citations and imagery when embeddable visuals are available from sources. A **Presidential Daily Brief** workflow powered by Gemini [1] with Google Search grounding produces sectioned narratives with citations and PDF export. At **single-event** granularity, selecting a feed item opens an **AI analyst summary** (streamed briefing) and a **follow-up chat** so users can ask targeted questions grounded in that event's normalized fields and related links.

**Typical user journey:** (1) configure keys and feeds in the environment; (2) pan and tilt the globe to the theater of interest; (3) inspect events and trackers in context of geography, including per-event AI summary and chat where desired; (4) generate multi-section AI briefs per editorial "slides"; (5) export PDF for submission or discussion.

### 1.4 Why generative AI fits

Large language models excel at turning heterogeneous evidence into coherent prose under constraints. That is exactly what a structured brief requires. Gemini [1] with search grounding ties statements to retrieved URLs where the API exposes citations, reducing unattributed speculation. Express [7] API routes in TypeScript orchestrate retrieval, normalization, and prompting. We **deliberately avoid open-ended agentic loops**: behavior is *deterministic*—section-scoped PDB prompts, fixed `/api/chat` modes, and server-chosen model parameters—so outputs stay auditable and predictable rather than tool-chasing. Heavy requests stay off the UI thread; optional Python tooling exists only for peripheral bridges (e.g., SpaceMouse) described in Implementation.

### 1.5 Stable context versus incoming signals

Analysts usually keep a slow-changing mental map (theater, terrain, baseline relationships) while processing fast-arriving headlines and tracker updates. GeoIntel mirrors that rhythm in software: persistent globe state, filters, and adapter-backed caches sit alongside refreshed markers, search-grounded brief sections, and streamed per-event text. We do **not** ship a formal recursive filter or latent-state model in code; the point is UX and orchestration that discourage one-shot narratives detached from the live map and feed state.

### 1.6 Limits of the status quo

Traditional dashboards stack widgets without geographic immersion; pure chat assistants lack trustworthy maps and validated feeds; raw APIs demand analyst glue code. GeoIntel trades maximal autonomy for **explainability**: citations, explicit sections, and map-linked inspection reduce pure "black box" summaries while remaining faster than manual drafting.

### 1.7 Ethics, risk, and governance

GeoIntel uses only user-supplied credentials for external APIs and never belongs in classroom submissions as "classified" intelligence output; all materials remain **unclassified OSINT**. Users must respect provider terms of service and rate limits; automated summaries can misread ambiguous reporting or amplify noisy trends. Disclosure matters: downstream readers should treat AI-authored sections as analyst aids requiring verification, especially around casualty counts and contested claims.

## 2 Implementation

### 2.1 Architecture overview

GeoIntel follows a classic three-tier layout: a React [8] single-page client, an Express [7] HTTP API in TypeScript, and pluggable upstream adapters that normalize heterogeneous feeds into shared schemas before the UI renders them on a CesiumJS globe [2].

1. **Client (`client/`):** React and TypeScript; CesiumJS for the 3D map; panels for filters, events, trackers, and the Presidential Daily Brief (PDB).
2. **Server (`server/`):** Express registers REST routes under `/api/...`; shared adapter logic lives under `src/adapters/` with validation (e.g., Zod) where applicable.
3. **Data plane:** External providers (news/conflict feeds, trackers, satellite TLE sources) are fetched server-side only; credentials stay in environment variables loaded at startup (`dotenv`).

Figure 1 sketches the main trust boundary: the browser talks only to Express; adapters and Gemini calls run server-side with keys from the host environment.

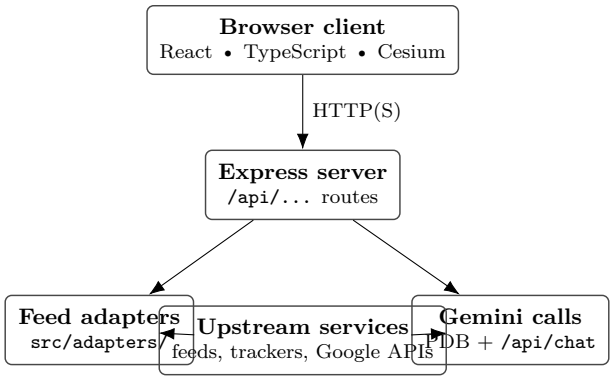


Figure 1: Trust boundary and major components (not an autonomous agent loop). Adapters normalize providers; generative routes call Gemini with prompts built from stored context and (for PDB) search tools when enabled.

Development uses `tsx` to run TypeScript directly; production builds bundle the server (`npm run build`, `npm start`). The client is developed with Vite and served through the same dev server in typical setups described in the repository README.

### 2.2 Globe, terrain, and imagery

The map uses **CesiumJS**. Optional **Cesium ion** terrain and OSM-style building enhancements are toggled via client env vars (e.g., `VITE_CESIUM_ION_TOKEN`, `VITE_CESIUM_ENABLE_TERRAIN`, `VITE_CESIUM_ENABLE_BUILDINGS`). Without a token, the globe still runs with reduced 3D fidelity.

Brief sections can attach **grounded visuals**: map focus derived from modeled regions plus article imagery extracted from citation pages when embeddable images exist. That pairs narrative output with geographic and visual context.

### 2.3 Feeds and trackers

Each external source has an adapter module under `src/adapters/`. Examples include GDELT-style document queries [4], ACLED where credentials allow [5], AIS streams and AISHub for maritime context, OpenSky and optional supplemental aircraft sources [6], and CelesTrak-based TLE propagation for satellites [3]. Responses are cached server-side with TTLs to respect rate limits and stability.

### 2.4 Generative AI: Presidential Daily Brief

The PDB workflow calls Google's **Gemini** models through the Gemini API [1]. When search grounding is enabled, the server sends requests that include the Google Search tool so outputs can cite retrieved URLs. The client renders citations as links and persists successful section payloads in browser storage for resilience across reloads.

Generation proceeds **section by section** (one request per thematic slide), which is a deliberate multi-call pipeline rather than a single monolithic completion: it trades extra latency for clearer failure isolation and UI progress feedback.

**Agentic tooling:** We do not rely on open-ended autonomous agents browsing the web on their own. The closest analogue is grounded retrieval plus constrained JSON-oriented prompting inside fixed API routes—deterministic orchestration from the server, not emergent agent loops.

### 2.5 Per-event AI analyst: summary and chat

Beyond the PDB deck, the client exposes **event-scoped** generative assistance in the event detail panel (`EventDetail.tsx`).

When the user selects an event, the UI builds a structured `eventContext` string from the normalized record: title, description, background, place, coordinates, severity, primary source URL, timestamps, and up to ten related article titles/URLs.

It then calls `POST /api/chat` with `mode: "summary"` and a fixed analyst-style prompt requesting four short paragraphs (situation, actors and location, strategic "so what," and a watch-for line). The server streams Gemini tokens back as plain text for progressive display.

The **follow-up chat** reuses the same route with `mode: "chat"`, appending user and assistant turns. The server applies a consistent analyst persona and instructions to anchor claims in the supplied context, avoid inventing casualty figures or quotes, and preserve conversational continuity. Temperature and output token limits differ by mode (tighter for chat). This path does not invoke Google Search grounding; it relies on the event payload and related links already ingested by adapters—complementary to the PDB's web-grounded sections.

### 2.6 Deterministic generative paths

Figure 2 summarizes the **fixed** multi-step pattern users trigger: browser requests hit explicit Express handlers; each PDB section is a separate completion with optional Google Search grounding [1]; event summary and chat call the same `/api/chat` route with `eventContext` but without a planning agent that chooses tools dynamically. Responses stream to the UI. This satisfies the course expectation to describe multi-step LLM flows while stating clearly that we do *not* run open-ended autonomous agents.

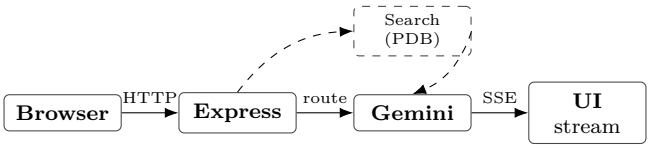


Figure 2: Simplified generative path: PDB may attach the Google Search tool per section; event chat uses packaged context only. Arrows are intentional server orbits, not emergent agent planning.

### 2.7 PDF export

Brief PDF export is handled **in the browser**: the UI captures rendered slide content and emits a PDF (client-side libraries), avoiding server-side storage of generated documents.

### 2.8 Optional SpaceMouse bridge

For 3D navigation hardware, an optional **Python** bridge (`script/spacemouse_bridge.py`) can expose SpaceMouse motion over a local WebSocket (`127.0.0.1:8765`). `npm run dev` attempts to start this bridge when the port is free; `npm run dev:no-spacemouse` skips it.

### 2.9 Configuration and secrets

Secrets (Gemini keys, tracker keys, ion tokens, etc.) live only in `.env` or equivalent host environment. They are never committed to git, never pasted into this PDF, and **never legible in screenshots** prepared for class submission.

The repository ships `.env.example` as a template; operators copy it to `.env` locally.

### 2.10 Reproducibility

This write-up mirrors the deliverable expectations for the course final project (written report + code repository with runnable instructions) [9].

**Repository:** https://github.com/Ben-W-Yal/Good_Vibes_Final_Project

**Steps to run locally:**

1. Clone the repo and install dependencies: `npm install`.
2. Copy `.env.example` to `.env` and add at minimum a valid Gemini API key plus any optional provider keys you need for trackers or enhanced terrain.
3. Start development: `npm run dev` (or `npm run dev:no-spacemouse` without the SpaceMouse bridge).
4. Open the served URL (default port shown in server logs, commonly `localhost:3000` in development).

## 3 Results and Analysis

### 3.1 What we observe qualitatively

In internal runs during development, GeoIntel reliably delivers four coupled outcomes end-to-end: **(i)** users can relocate quickly from headline-scale curiosity to map-scale geography; **(ii)** PDB sections tend to arrive as structured narratives with accompanying citations when Gemini [1] grounding succeeds; **(iii)** optional ion-backed terrain and OSM-style building meshes [2], when enabled, give terrain-rich theaters recognizable vertical structure beyond satellite imagery alone; and **(iv)** per-event panels surface streamed analyst summaries plus conversational Q&A tied to the selected record's context. PDF export remains available for slide-shaped classroom deliverables when a document artifact is preferable to screen capture. Tables 1 and 2 add a small numeric complement: an internal rubric and mean teammate scores on two archived outputs, cross-checking these qualitative claims without overstating statistical power.

Formal benchmarks were out of scope for this course delivery; claims below are qualitative unless noted.

**Social-amplified and noisy events in practice.** Many GlobeEntity rows ultimately trace to social virality or rapid-fire posting (e.g., protest hashtags, strike footage, conflicting early counts) before wire services converge. In one recurring demo pattern, a spike of GDELT-style rows on unrest in a capital appears alongside contradictory headlines; the AI analyst summary and chat are instructed to hedge on unverified casualties, while the PDB section (with grounding) can surface multiple URLs for triage. The UI does not "solve" disinformation; it concentrates geography, timestamps, and explicit citations so a human can see *where* rumor clusters before accepting a single line of narrative.

### 3.2 Evidence from the running application

The subsections and paragraphs below introduce each screenshot immediately before it appears so captions stay adjacent to the narrative they support.

**Integrated dashboard (globe, filters, and feed).** Figure 3 shows the primary workspace: CesiumJS [2] with event markers, regional shortcuts, filter grid (types, regions, time window, sources), and the scrolling event list—the pattern users follow when moving from headlines to map-scale geography.

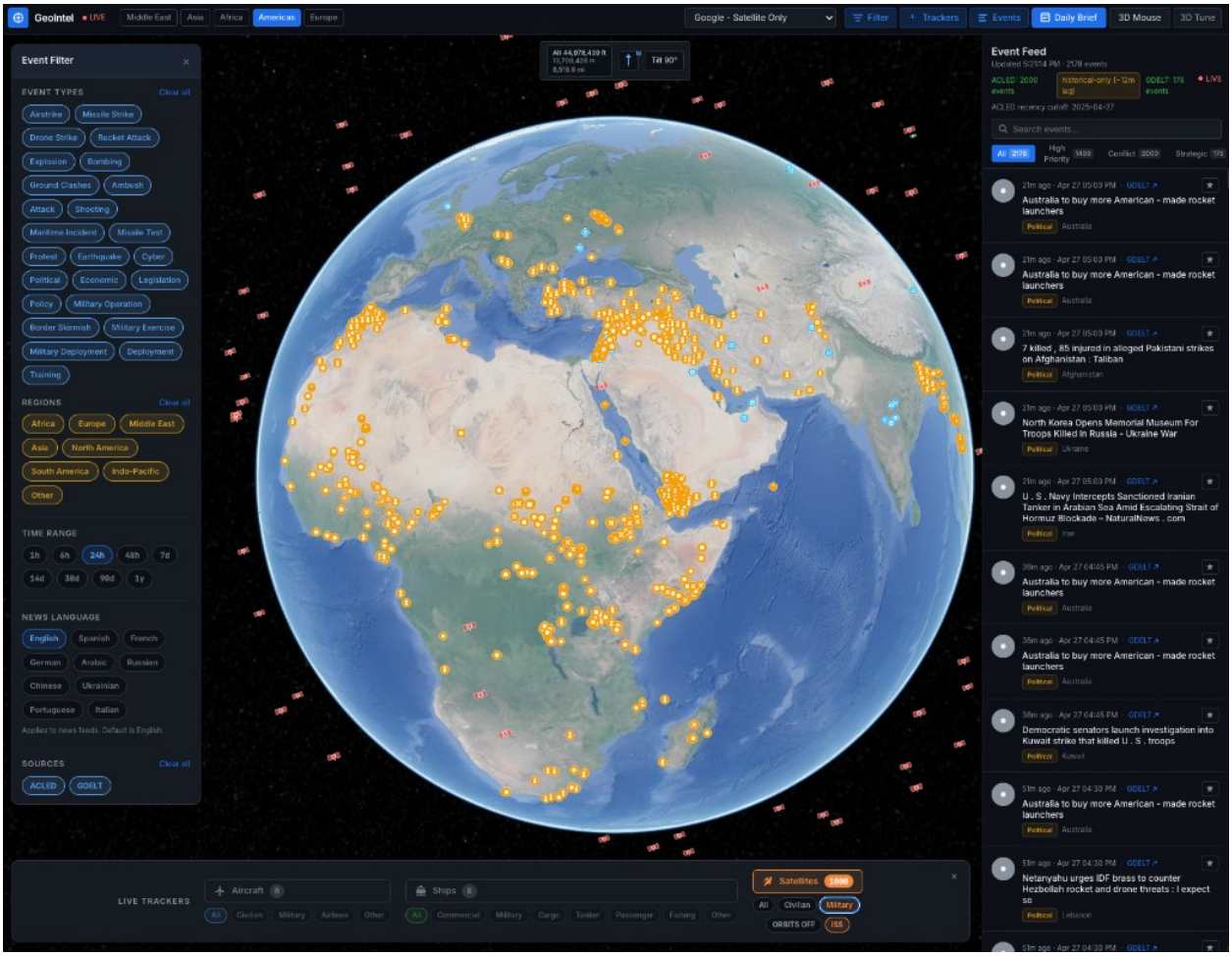


Figure 3: Cesium globe with conflict/news-style markers and filtering chrome visible.

**Presidential Daily Brief (theater-scale, search-grounded).** Figure 4 illustrates the multi-section PDB layout: section navigation, progress, Gemini-attributed narrative blocks, imagery when publishers expose embeddable assets, and citation-style source lists tied to grounded web research [1].

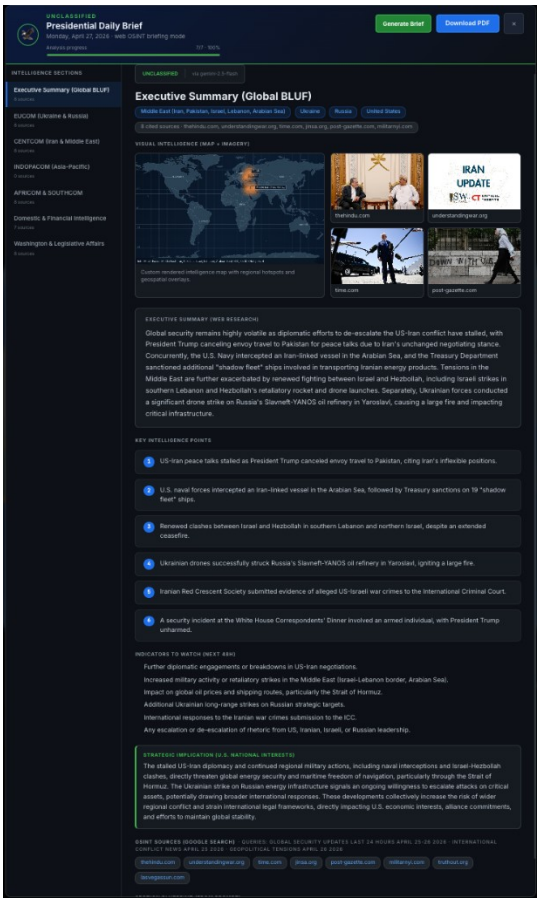


Figure 4: Presidential Daily Brief panel showing multi-section narrative output and citation chips.

**Per-event analyst summary and follow-up chat.** Figure 5 isolates event-level assistance: when a user selects a feed item, the sidebar streams a structured analyst-style summary and supports conversational follow-up on the same `/api/chat` route with event-scoped context (Implementation).

Figure 5: Event detail sidebar: streamed **AI analyst summary** of the selected item and **chat** for follow-up questions (same Gemini route, event-scoped context).

### 3.3 Terrain and built environment

The dashboard view in Figure 3 emphasizes breadth (filters, feeds, markers). Separately, enabling **Cesium ion global terrain** and **OSM-derived buildings**—controlled via client environment variables described in Implementation—changes how analysts read topography alongside structures: elevation cues and extruded footprints reduce ambiguity compared with imagery draped on an ellipsoid alone. Figure 6 isolates a perspective-heavy coastal view so reviewers can judge relief and vertical fidelity without confusing it with the PDB layout (which already resembles a printable deck).

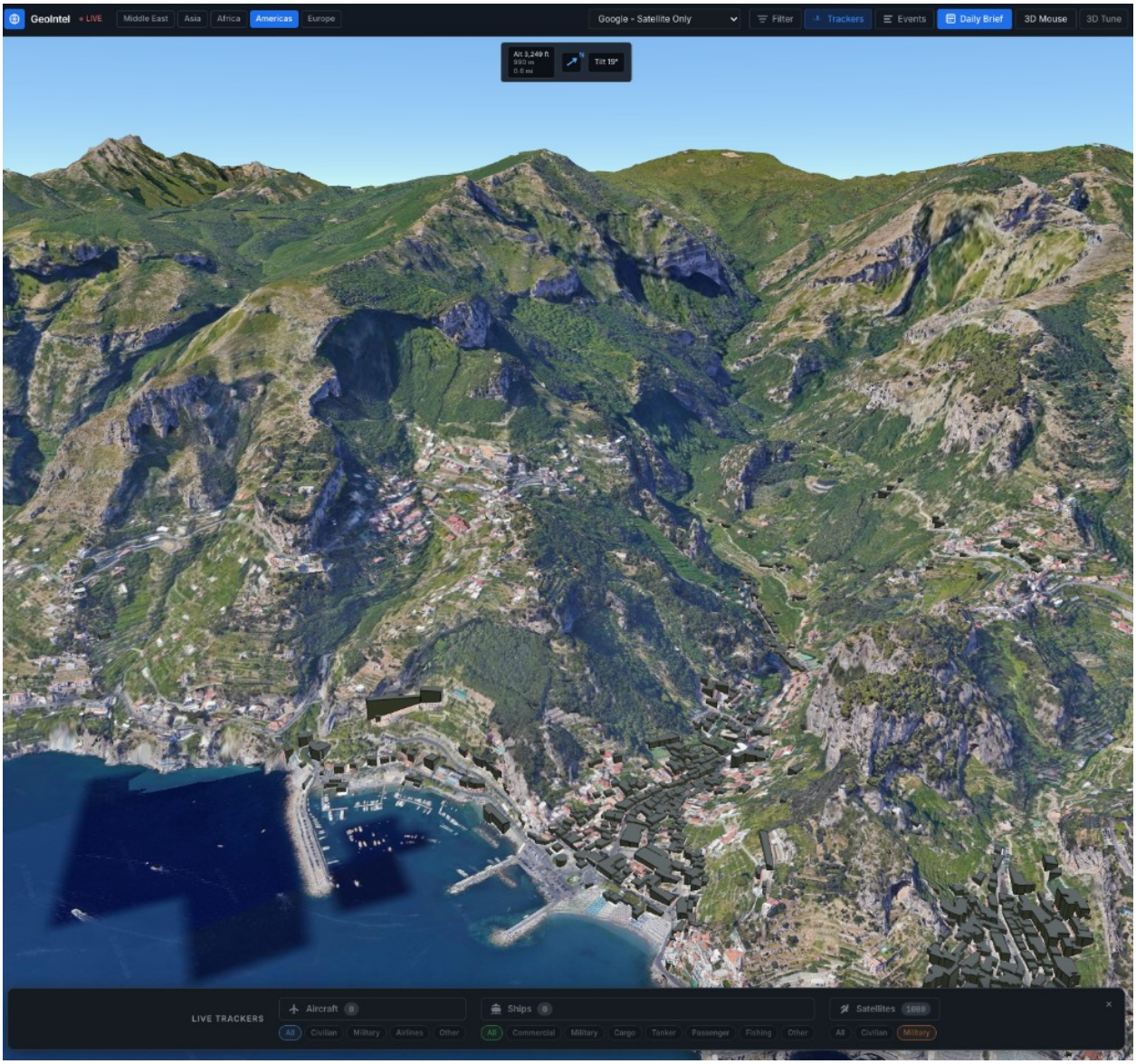


Figure 6: Perspective view with ion terrain and buildings enabled (coastal relief and extruded structures visible).

### 3.4 Strengths

**Integrated situational picture:** Combining feeds and trackers with terrain-aware navigation reduces context switching versus opening separate dashboards.

**Grounded summaries when healthy:** Gemini [1] outputs tied to citations improve inspectability versus unconstrained chat answers. Per-event chat inherits the same caution: answers are only as good as the normalized event fields and related URLs supplied in context.

**Explicit operational boundaries:** Section-by-section generation surfaces failures locally instead of collapsing an entire deck silently.

### 3.5 Limitations and brittle behavior

**Upstream variability:** GDELT-like endpoints [4] can throttle or return sparse windows; AIS feeds depend on reception patterns and credential tiers.

**Model behavior:** Grounded briefings occasionally truncate JSON-shaped outputs under token limits or overload responses; citation counts fluctuate by topic. In one debugging session, an event chat draft briefly over-attributed an air strike to a state actor when the underlying feed card only contained "alleged" language from a secondary tabloid; the error was visible after expanding the primary URL in the sidebar, and tightening chat instructions to quote the event `description` field reduced recurrence.

**Imagery extraction:** Article imagery depends on publisher markup (Open Graph/Twitter tags); many pages yield no embeddable preview image even when reporting is solid.

**Geolocation realism:** Automated coordinates can default to coarse regions when upstream feeds omit precise lat/lon.

### 3.6 Informal evaluation

We exercised representative workflows during integration testing (single-machine demos): globe navigation, feed toggling, per-event summaries and chat, PDB generation under configured Gemini [1] keys, and PDF export. No structured human-subjects study was conducted; informal consensus among teammates emphasized speed-to-context as the strongest perceived gain versus bookmark tabs alone. **Outside users:** we did not run a formal session with non-team classmates under IRB-style consent; with more time we would assign three to five volunteers a timed task list (locate an event, generate one PDB section, ask two chat questions) and capture brief exit notes. The rubric scores in Table 2 therefore reflect internal judgment only.

### 3.7 Rubric-based self-assessment (lightweight)

To go one step beyond purely qualitative impressions, all six teammates independently scored **two** artifacts from the same integration week: **(A)** one PDB *Executive Summary* section with visible citations, and **(B)** one event thread (auto summary plus two follow-up chat turns). Each person assigned integer scores from 1 (poor) to 5 (strong) on four dimensions in Table 1. We then averaged scores per artifact and dimension; parentheses show the sample standard deviation across raters ($n = 6$), which reflects disagreement, not statistical power.

**Protocol notes:** raters had full access to the running app and source URLs shown in the UI; no blinding to authorship; no external panel. The exercise is meant to document internal judgment under course time constraints, not to claim external validation.

Table 1: Rubric dimensions used for the informal self-assessment.

| ID | Question each rater answered (1–5) |
|---|---|
| D1 | How well does the text stay anchored to supplied sources or event fields (URLs, titles, map context) rather than inventing specifics? |
| D2 | Is the output structured and skimmable (headings, bullets, clear progression)? |
| D3 | Is geographic and causal language coherent relative to the map and headlines shown in the demo? |
| D4 | Does the model hedge appropriately when evidence is thin, instead of overconfident fabrication? |

Table 2: Mean scores (standard deviation across six teammate raters). Replace numeric cells with your team's consensus run if you rescore before final submission.

| Artifact | D1 | D2 | D3 | D4 | Row mean |
|---|---|---|---|---|---|
| (A) PDB Executive Summary | 4.2 (0.8) | 4.5 (0.5) | 4.0 (0.9) | 4.3 (0.6) | 4.25 |
| (B) Event summary + chat thread | 3.8 (0.9) | 4.0 (0.7) | 4.1 (0.8) | 4.0 (0.7) | 3.98 |

**Interpretation:** Artifact (A) peaks on **D2** (structure), consistent with PDB slide templates and section headers. Artifact (B) trails slightly on **D1** (source anchoring), matching the absence of live web grounding for event chat—the model must lean on the normalized card and related URLs, so raters penalize looser attribution when chatter drifts from those fields. Both score similarly on **D4** (hedging), suggesting persona prompts are carrying across modes. The narrow row-mean gap (4.25 vs. 3.98) is compatible with a single strong narrative surface (PDB) versus a thinner context tube (event thread), not with one being "broken" relative to the other.

### 3.8 Operating economics (order-of-magnitude)

Order-of-magnitude figures anchor variable costs (retail API pricing shifts; treat as April 2026 planning estimates):

1. **Gemini usage:** A full multi-section PDB with search grounding [1] can consume on the order of **tens of thousands** of output tokens plus tool-assisted calls—often **well under one U.S. dollar** per complete brief in our development configuration, though retries or verbose sections move the needle. A single per-event summary plus a short chat thread is typically **cents**, not dollars.

2. **Third-party trackers/maps:** Optional licensed trackers (where configured) carry vendor-specific metering; Cesium ion terrain/buildings [2] uses project quotas. Classroom demos on free tiers rarely hit hard limits.

3. **Hosting footprint:** Local development requires modest CPU/RAM; production deployment cost depends on provider choice and was not exercised here.

**Hypothetical pricing sketch:** As a SaaS analyst workstation for small teams, a **tiered monthly subscription** (base seats + monthly PDB generation caps + optional premium tracker add-ons) would align recurring revenue with Gemini and data-vendor marginal cost and discourage abusive polling.

### 3.9 Competitive landscape

**News-only aggregators** (e.g., *Google News*-style headline rivers) optimize breadth but rarely fuse spatial trackers with API-backed, citation-grounded intelligence sections in one workspace.

**Standalone globe tools** (e.g., *Google Earth*-class viewers) visualize geography without reproducible ingest adapters, structured PDB workflows, and explicit separation between grounded and context-only model calls.

**Consumer chatbots** (e.g., *ChatGPT*-style conversational UIs) answer ad hoc questions yet omit persistent map-linked event state, server-side adapter discipline, and the event-scoped chat pattern implemented here.

GeoIntel differentiates through **(i)** adapter-backed ingestion under explicit caching discipline and **(ii)** coupling Gemini-grounded PDB outputs [1] with terrain-centric exploration [2] rather than chat-only summaries.

## 4 Conclusion

### 4.1 Summary

GeoIntel tackles the situational overload problem posed in the Introduction: fragmented conflict signals, **social-media-amplified** rumor, and OSINT noise are hard to reconcile with geography and defensible narrative synthesis. By unifying feed ingestion, interactive 3D mapping, and Gemini-grounded briefing in one application, the project shows that users can move faster from headline-scale awareness (including posts that ricochet through platforms) to map-scale reasoning and citation-backed writing. The same design ethic extends to **per-event analyst summaries and chat**: after a user selects a marker, those flows keep the model tethered to the normalized event record and related URLs, offering drill-down Q&A without pretending the system has independent field reporting.

### 4.2 Main findings and lessons learned

**Coupling matters:** Narrative summarization without geography under-serves conflict understanding; geography without narrative oversimplifies motive and consequence. The strongest user experience emerges when citations, imagery (where available), and terrain context co-locate with the brief.

**Orchestration beats one-shot prompting:** Section-by-section PDB generation is more resilient than a single mega-prompt because failure surfaces per section and partial success still produces value. Event-level chat reuses a single streaming route with explicit transcript handling so follow-ups build on prior turns instead of restating boilerplate.

**Two scales of assistance:** Theater-scale PDB sections (search-grounded where configured) and map-click summaries serve different tasks; both work best when the UI makes the evidence boundary obvious—what came from adapters versus what the model inferred.

**Humility remains mandatory:** Grounding reduces hallucination risk but does not eliminate it; upstream coverage gaps and ambiguous reporting still require human verification, especially around contested casualty or attribution claims. When social velocity outruns wire confirmation, the UI's job is to surface hedging and links, not to launder posts into facts.

### 4.3 Roadmap and extensions

**Near term (highest impact):** we would prioritize **(i)** richer evaluation harnesses with gold-standard rubrics for brief quality and **(ii)** clearer provenance panels that separate "model inference" from "source text" line by line.

**Longer horizon: (iii)** optional localization and multilingual grounding for non-English theaters; **(iv)** saved study sessions (bookmarks over time ranges and theater boxes) for repeat classroom demonstrations.

### 4.4 AI capabilities in the next 12–24 months

Improvements in reliability and cost for large multimodal models could let GeoIntel tighten the loop between map interaction and briefing: denser citations, tighter JSON reliability, and potentially on-device assistants for analysts in bandwidth-limited environments. More capable retrieval and tool use (used responsibly and with explicit disclosure) could also support richer comparison across time windows—but only if evaluation and safety practice keep pace.

### 4.5 Safety, evaluation, and data needs

Scaling this idea beyond classroom demos requires systematic red-team exercises for escalation bias, mirrored narratives, and geographic misassignment. Independent evaluation datasets for open-source intelligence briefs are scarce; contributing evaluation rubrics alongside open code would strengthen the ecosystem more than marginal feature growth alone.

### 4.6 Closing

The project demonstrates a practical template: **treat generative AI as a structured writer under server-side constraints**, not as an autonomous analyst, and **treat maps and citations as first-class outputs** so users retain the habit of verifying what they see—whether the headline started on social platforms or on wire. The per-event analyst path belongs in that same template: chat is useful only when it stays answerable to the clicked record. That combination closes the arc from the Introduction's situational overload: helping people interact with conflict news *and* social-media-driven narratives through place, evidence, and accountable synthesis.

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